

Urban Land Cover Analysis Pipeline

Digital Image Processing — Project Proposal | Group H | Nisarg Shah, Ansar Rahman | September 18, 2026

1. Image / Dataset Information

Application domain: Satellite / aerial remote sensing (urban land cover).

Source: NAIP (National Agriculture Imagery Program) 1 m aerial orthoimagery over downtown/central Houston, TX, retrieved from the USGS National Map imagery service (imagery.nationalmap.gov/.../USGSNAIPImagery), the same NAIP holdings indexed for browsing/ordering via USGS EarthExplorer (<https://earthexplorer.usgs.gov>). NASA Worldview (<https://worldview.earthdata.nasa.gov>) will be used for wide-area context browsing only.

License: NAIP imagery is produced by the USDA Farm Production and Conservation Business Center and is U.S. Government public-domain data (free to use/redistribute); USDA/USGS attribution will be included in the final report.

Description: A true-color aerial orthophoto covering a mixed urban scene (downtown Houston and its I-45/I-59/US-59 highway interchange) that contains, within one frame, dense high-rise buildings, a multi-level highway interchange, a regular street grid, vegetation, and the Buffalo Bayou. At NAIP's native 1 m/pixel resolution, individual roads, building footprints, and interchange ramps are clearly resolvable — unlike coarser 30 m Landsat imagery, which was evaluated and rejected for this project because it cannot resolve city-street-scale structure. **A 1024×1024 px sample crop was pulled from the live NAIP service during proposal preparation and run through every stage of the pipeline below (gamma/stretch, CLAHE, Laplacian/median, FFT low/high-pass, Canny, Harris) to confirm the plan executes end-to-end on real data and produces interpretable before/after output** — e.g., Canny recovers the interchange ramps and street grid as clean line structure, and Harris marks building corners and ramp junctions.

2. Project Objective

The objective is to build a classical image-processing pipeline that improves the visual and structural interpretability of an urban satellite scene for land-cover discrimination — i.e., to make roads, buildings, vegetation, and water easier to separate and measure without machine learning. Concretely, the project will: (a) correct illumination and contrast so that shadowed or hazy regions are usable, (b) enhance local contrast across radiometrically different land-cover classes, (c) sharpen and denoise structural detail, (d) separate coarse land-cover regions from fine structural detail in the frequency domain, and (e) extract edges and corner-like features (road intersections, building outlines) that would support downstream segmentation.

This scene was chosen because it is a single image with genuinely heterogeneous content, which lets each technique be evaluated against a distinct sub-problem (e.g., contrast correction on shadowed rooftops vs. edge detection on road grids) rather than being applied generically. Free, license-clear availability and high spatial variety also make it well suited to demonstrating clear, checkable before/after results.

3. Planned Image-Processing Pipeline

Point Processing

- **Gamma correction** — brighten shadowed/underexposed regions (buildings' shadow sides, tree canopy interiors) by applying an $\text{output} = \text{input}^{(1/\gamma)}$ transform with γ tuned to the scene's mean shadow intensity, rather than a fixed value.
- **Linear contrast stretching** — expand the dynamic range of low-contrast areas such as parking lots and light rooftops that cluster in a narrow intensity band, using the scene's actual min/max (or a percentile clip) rather than assuming 0–255 is fully used.

Histogram Processing

- **CLAHE (Contrast-Limited Adaptive Histogram Equalization)** — equalize contrast locally, tile by tile, so that both dark vegetation and bright water/soil regions gain contrast simultaneously — something global histogram equalization would over- or under-correct, since it applies one mapping to the whole heterogeneous scene.

Spatial Filtering

- **Laplacian sharpening** — boost high-frequency edges of roads and building boundaries that CLAHE may soften, using an unsharp-mask-style addition of the Laplacian response scaled by a weight chosen to avoid overshoot/ringing.

- **Median filtering** — suppress salt-and-pepper-like sensor noise and small texture speckle in vegetation regions while preserving edges better than a mean filter, with kernel size chosen as the smallest that visibly reduces speckle without blurring road edges.

Frequency-Domain Filtering

- **FFT low-pass filtering** — isolate broad land-cover regions and suppress high-frequency sensor/compression artifacts, with the cutoff radius chosen from the image's radial power spectrum rather than an arbitrary value.
- **FFT high-pass filtering** — recover fine structural detail (street grids, building rooftops) that the low-pass step removes, illustrating the trade-off between the two and motivating a band-selective approach if time allows.

Edge & Corner Detection

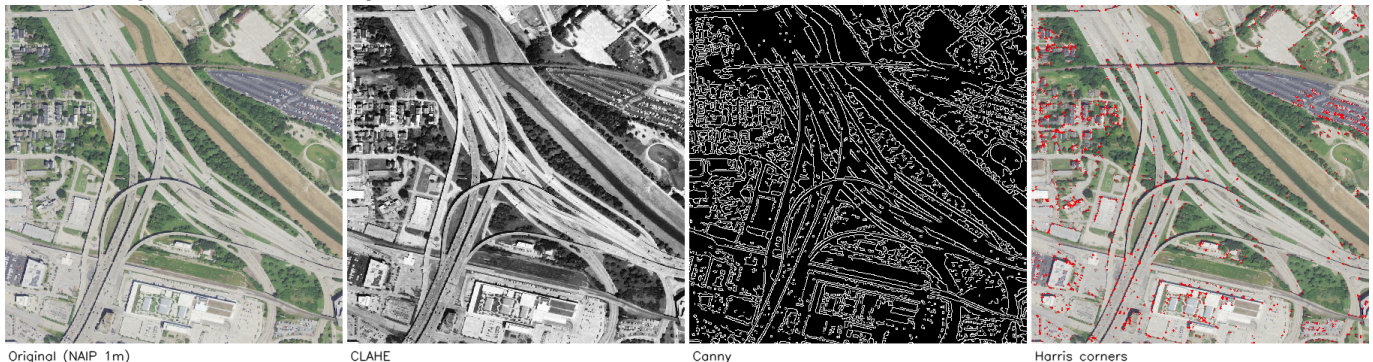
- **Canny edge detection** — outline building footprints and road boundaries; the two hysteresis thresholds will be set from the image gradient-magnitude histogram (not defaults) and justified against false-edge vs. missed-edge trade-offs.
- **Harris corner detection** — identify road intersections and building corners as candidate keypoints, useful as an interpretable proxy for structural complexity and as a natural bridge to the optional detection extension below.

This plan covers all five required categories (Point Processing, Histogram Processing, Spatial Filtering, Frequency-Domain Filtering, and Edge & Corner Detection). As noted in the assignment, exact parameters and the final subset of techniques may be adjusted after initial experimentation.

4. Optional Deep-Learning Extension

If pursued, a pretrained YOLO (e.g., YOLOv8) object detector will be run on a cropped patch of the 1 m NAIP scene to attempt vehicle/building detection. The discussion will focus on where detection succeeds versus fails at satellite/aerial scale — e.g., degraded performance on small, low-contrast, or densely packed objects — and how that failure mode relates to the classical edge/corner results obtained earlier in the pipeline.

Feasibility Check: Real Output on a Live NAIP Sample



Crop of the tested 1024×1024 NAIP scene (I-45/I-59 interchange, Buffalo Bayou). CLAHE recovers building and lot detail lost in flat pavement/roof tones; Canny cleanly traces ramp curves and street edges; Harris concentrates corner responses on buildings, parking-lot markings, and ramp junctions rather than firing uniformly on noise — confirming these techniques are meaningful at this resolution, not just executable.

Note: the pipeline above was smoke-tested end-to-end on a real 1024×1024 NAIP crop of this scene (gamma $\gamma=1.4$; CLAHE clip=2.5, tile=8×8; median k=5; FFT low-pass radius=60 px; Canny thresholds from the median-intensity heuristic; Harris block=2, k=0.04) — all stages ran without error and produced non-degenerate, visually interpretable output. These exact values are starting points, not final settings; final values will be tuned and justified quantitatively in the project submission.